Faculty Vita: Brian P. Chaplin

1. Education

B.S. Civil Engineering, University of Minnesota, 1999
M.S. Civil Engineering, University of Minnesota, 2003
Ph.D. Environmental Engineering, University of Illinois at Urbana-Champaign, 2007

2. Academic Experience

Department of Chemical Engineering, University of Illinois at Chicago
Director of Graduate Studies, 2024–present. Full-Time
Department of Chemical Engineering, University of Illinois at Chicago
Professor, 2022–present. Full-Time
Department of Chemical Engineering, University of Illinois at Chicago
Associate Professor, 2018–2022. Full-Time
Department of Chemical Engineering, University of Illinois at Chicago
Assistant Professor, 2013–2018. Full-Time
Department of Civil and Environmental Engineering, Villanova University
Assistant Professor, 2010–2013. Full-Time
Department of Chemical and Environmental Engineering, University of Arizona,
Postdoctoral Research Associate, 2008–2010. Full-Time
Department of Civil and Environmental Engineering, University of Illinois at Urbana-Champaign, Postdoctoral Research Associate, 2007-2008. Full-Time

3. Non-Academic Experience

U.S. Geological Survey, Mounds View, MN, Hydrologist, 1999-2003. Full-Time

4. Certifications or professional registrations

E.I.T. certified (Minnesota since 1999)

5. Current membership in professional organizations

American Institute of Chemical Engineers
Association of Environmental Engineering and Science Professors

6. Selected Honors and awards

- 2019 *Environmental Science & Technology* and *Environmental Science & Technology Letters Virtual Issue: Early Career Scientists*

- 2018 *Environmental Science & Technology Best Paper Award*
- *NSF Early CAREER Development Award* (2015-2020)
- *Environmental Science & Technology Excellence in Review* Award (2017)
- *Environmental Science & Technology Letters Excellence in Review* Award (2016)
- UIC College of Engineering Research Award (2015)
- Summer Research Fellowship/Research Support Grant, Villanova University (2011)
- Environmental Water Resources Institute's 2011 Samuel Arnold Greeley Award (Best Paper). **Chaplin** et al., (2009). *Journal of Environmental Engineering-ASCE.* **135(8), 666-676.**

7. Selected service activities

- Associate Editor, *ACS ES&T Engineering*, 2024-present
- Director of Graduate Studies, Department of Chemical Engineering, 2024-present
- Editorial Advisory Board, *Environmental Science and Technology*, 2020-2024
- Member, *Chemical Engineering Departmental Advisory Board*, Department of Chemical Engineering 2018-present
- Chair, *Safety Committee*, Department of Chemical Engineering, 2018-present
- Chemical Engineering Department Seminar Series Coordinator, 2018-present.
- NSF CAREER panel reviewer, CBET-Env. Eng. Program, Nov 2015, Sept 2016
- NSF panel reviewer, CBET-Env. Eng. Program, March 2016, February 2017

8. Selected Publications from the last 5 years

1. Raghavan, S.; **Chaplin, B. P.**; Mehraeen, S. Elucidating the Impact of the Electrocatalyst Surface and Applied Potential on the Decarboxylation of the PFOA Radical on the Ti_4O_7 Facet Using DFT Simulations. *J. Phys. Chem. C* **2025**, *129* (23), 10234–10248. DOI: 10.1021/acs.jpcc.5c01234.
2. Zheng, X.; Gupta, N.; He, H.; Bindra, J. K.; Hossain, S.; Vizuet, J. P.; Nadeali, A.; Zangeneh, D.; Singh, R. P.; Klie, R. F.; **Chaplin, B. P.**; Brezinsky, K.; Poluektov, O. G.; Niklas, J.; Zapol, P.; Glusac, K. D. A Light-Responsive Metal–Organic Framework with Perchlorinated Nanographene Ligands. *J. Am. Chem. Soc.* **2025**, *147* (19), 16420–16428. DOI: 10.1021/jacs.5c02844.
3. Saffar-Avval, S.; Modiri Gharehveran, M.; Alvarez Ruiz, R.; Lee, L. S.; **Chaplin, B. P.** Matrix Effects on Electrochemical Oxidation of Per- and

Polyfluoroalkyl Substances in Sludge Centrate. *Environ. Sci. Technol.* **2025**, *59* (16), 8263–8273. DOI: 10.1021/acs.est.4c13720.

4. Mohamed, M. S.; **Chaplin, B. P.**; Abokifa, A. A. Screening of Transition Metals for PFAS Adsorption: A Comparative DFT Investigation. *Chem. Eng. Sci.* **2025**, *307*, 121363. DOI: 10.1016/j.ces.2025.121363.
5. Khalid, Y. S.; Islam, S. M. M.; Mehraeen, S.; **Chaplin, B. P.** Reactive-Transport Modeling of Oxidation Pathways of Insensitive High Munitions in Porous Flow-Through Electrodes. *ACS ES&T Eng.* **2025**, DOI: 10.1021/acsestengg.4c00895.
6. Mohamed, M. S.; **Chaplin, B. P.**; Abokifa, A. A. Ab Initio Investigation of Per- and Poly-fluoroalkyl Substances (PFAS) Adsorption on Zerovalent Iron (Fe^0). *ACS Omega* **2024**, *9* (44), 44532–44541.
7. Hafeez, S.; Khanam, A.; Cao, H.; **Chaplin, B. P.**; Xu, W. Novel Conductive and Redox-Active Molecularly Imprinted Polymer for Direct Quantification of Perfluorooctanoic Acid. *Environ. Sci. Technol. Lett.* **2024**, *11* (8), 871–877.
8. Misal, S. N.; Li, D.; Kim, S.; **Chaplin, B. P.** Effect of Solution Conditions and Applied Potential on Ion Transport in TiO_2 Nanopores. *ACS ES&T Eng.* **2024**, *4* (10), 2495–2505.
9. King, J. F.; **Chaplin, B. P.** Electrochemical Reduction of Per- and Polyfluorinated Alkyl Substances (PFAS): Is it Possible? Applying Experimental and Quantum Mechanical Insights from the Reductive Defluorination Literature. *Curr. Opin. Chem. Eng.* **2024**, *44*, 101014.
10. Mohamed, M. S.; **Chaplin, B. P.**; Abokifa, A. A. Adsorption of Per- and Poly-fluoroalkyl Substances (PFAS) on Ni: A DFT Investigation. *Chemosphere* **2024**, *357*, 141849.
11. Islam, S. M. M.; **Chaplin, B. P.** Electrochemical Destruction of Insensitive High Explosives using Magneli Phase Titanium Oxide Reactive Electrochemical Membranes. *ACS ES&T Eng.* **2024**, *4* (5), 1240–1252.
12. Almassi, S.; Ren, C.; Dandu, N. K.; Ngo, A.; Liu, J.; **Chaplin, B. P.** Mechanistic Study of Electrocatalytic Perchlorate Reduction using an Oxorhenium Complex Supported on a Ti_4O_7 Support. *ACS Catal.* **2024**, *14*, 2597–2608.
13. Raghavan, S.; **Chaplin, B. P.**; Mehraeen, S. Small-Molecule Adsorption Energy Predictions for High-Throughput Screening of Electrocatalysts. *J. Chem. Inf. Model.* **2023**, *63* (17), 5529–5538.
14. Plunkett, S. T.; Kondori, A.; Chung, D. Y.; Wen, J.; Wolfman, M.; Lapidus, S. H.; Ren, Y.; Amine, R.; Amine, K.; Mane, A. U.; Asadi, M.; Al-Hallaj, S.; **Chaplin, B. P.**; Lau, K. C.; Wang, H.; Curtis, L. A. A New Cathode Material for a Li-O_2 Battery Based on Lithium Superoxide. *ACS Energy Lett.* **2022**, *7* (8), 2619–2626.

15. Almassi, S.; Ren, C.; Liu, J.; **Chaplin, B. P.** Electrocatalytic Perchlorate Reduction Using an Oxorhenium Complex Supported on a Ti_4O_7 Reactive Electrochemical Membrane. *Environ. Sci. Technol.* **2022**, *56* (5), 3267–3276.
16. Khalid, Y. S.; Misal, S. N.; Mehraeen, S.; **Chaplin, B. P.** Reactive-Transport Modeling of Electrochemical Oxidation of Perfluoroalkyl Substances in Porous Flow-through Electrodes. *ACS ES&T Eng.* **2022**, *2* (4), 713–725.
17. Samonte, P. R. V.; Li, Z.; Mao, J.; **Chaplin, B. P.**; Xu, W. Pyrogenic Carbon-Promoted Haloacetic Acid Decarboxylation to Trihalomethanes in Drinking Water. *Water Res.* **2022**, *210*, 117988.
18. Plunkett, S. T.; Zhang, C.; Lau, K. C.; Kephart, M. R.; Wen, J. G.; Chung, D. Y.; Phelan, D.; Ren, Y.; Amine, K.; Al-Hallaj, S.; **Chaplin, B. P.**; Wang, H.; Curtis, L. A. Electronic Properties of Ir_3Li and Ultra-nanocrystalline Lithium Superoxide. *Nano Energy* **2021**, *90*, 106549.

9. Professional development activities

Science Leadership Council, *Current,* Chicago, IL, 2017-present